Reg. No. | | | | | | | | | | | [17BS4MA02]

FOURTH SEMESTER B.TECH EXAMINATION, JUNE–2019

LINEAR ALGEBRA AND VECTOR SPACES

Time:3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full questions, choosing at least **TWO** questions from each Part.
2. Any missing data may be suitably assumed

PART-A

1. **a.** Show that the set $V=\{a+b\sqrt{2}\,/\,a,b\in Q\}$ where Q is the set of rationals is a vector space over the field F=Q under the usual addition and scalar multiplication. **08**

b. Find the row space and null space of the matrix $A=\begin{pmatrix}1 & -2 & -1 & 3 & 2\\ 2 & -2 & -3 & 6 & 1\\ -1 & -4 & 4 & -3 & 7\end{pmatrix}$. **06**

2. **a.** Solve system of equation $3x+2y+3z=-5$, $3x+y-3z=4$ and $-3x+4y+7z=-7$ by Cramer's rule. **04**

b. Show that the transformation $y_1=2x_1-2x_2-x_3$, $y_2=-4x_1+5x_2+3x_3$ and $y_3=x_1-x_2-x_3$ is regular and find the inverse transformation. **05**

c. The following is a description of a two-industry model: We have two industries, electric company E and water company W. Output for both companies is measured in dollars. The electric company uses both electricity and water (input) in the production of electricity (output), and the water company uses both electricity and water (input) in the production of water (output). Suppose that the production of each dollar's worth of electricity requires \$0.3 worth of electricity and \$0.1 worth of water, and the production of each dollar's worth of water requires \$0.2 worth of electricity and \$0.4 worth of water. The final demand from the outside sector of the economy is \$12 million for electricity and \$8 million for water. Question: How much electricity and water should be produced to meet this final demand? How can we find the internal demand of E and W? **05**

3. **a.** Solve the linear system of equations $x+y+z=9$, $x-2y+3z=8$ and $2x+y-z=3$ by using matrix method. **04**

b. Find all the solutions of the system of homogeneous equations $3x_1+4x_2-x_3-6x_4=0$, $2x_1+3x_2+2x_3-3x_4=0$, $2x_1+x_2-14x_3-9x_4=0$ and $x_1+3x_2+13x_3+3x_4=0$. **05**

c. Test for consistency and solve the system of the equations $x+2y+3z=14$, $4x+5y+7z=35$ and $3x+3y+4z=21$. **05**

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4. a. Solve $X' = \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix} X + \begin{bmatrix} 3e^t \\ e^t \end{bmatrix}$ by diagonalization & hence discuss the stability of the system. **08**

b. If $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$ is matrix then obtain the matrix D such that $P^{-1}AP$ is a diagonal matrix. **06**

PART-B

5. a. Reduce quadratic form to canonical form sum of square by orthogonal transformation Also find the nature, rank, index &signature $q(x) = 2x_1^2 + 2x_2^2 + 2x_3^2 + 2x_1x_3$. **08**

b. Find the singular values and orthonormal vectors for the matrix A= $\begin{bmatrix} -3 & 1 \\ 6 & -2 \\ 6 & -2 \end{bmatrix}$. **06**

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6. a. Define Cartesian product of two sets and Relation with a suitable example. **04**

b. Given $f: R \to R$ be defined by $f(x) = \begin{cases} 3x-5, \text{ if } x > 0 \\ -3x+1, \text{ if } x \le 0 \end{cases}$, determine $f\left(-\frac{5}{3}\right)$, $f^{-1}(3)$ and $f^{-1}[-5,5]$. **05**

c. If a function $f: R \to R$ defined by $f(x) = 2x+5$ and $g: R \to R$ defined by $g(x) = \frac{1}{2}(x-5)$ then prove that g is an inverse of f. **05**

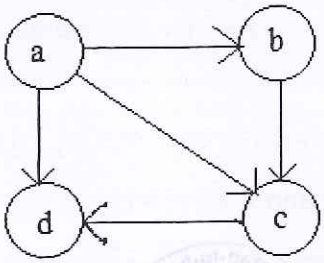
7. a. Given $A = \{1,2,3,4\}$ and R be a relation defined on A by xRy if and only if x divides y. Write down R as a set of ordered pairs. Draw the diagraph of R. Determine the in degree and out degree of vertex and also write M_R. **08**

b. Let $A = \{1,2,3,4\}$ determine the nature of the following relations on A:

i. $R_1 = \{(1,1),(2,2),(3,3),(4,4),(1,2),(2,1),(3,4),(4,3)\}$

ii. $R_2 = \{(1,2),(1,3),(3,1),(1,1),(3,3),(3,2),(1,4),(4,2),(3,4)\}$

iii. R_3 represented by the following diagraph: **06**



8.

a. Show that $\{u_1, u_2\}$ is an orthogonal set where $u_1 = \begin{bmatrix} -7 \\ 1 \\ 4 \end{bmatrix}$, $u_2 = \begin{bmatrix} -1 \\ 1 \\ -2 \end{bmatrix}$ and express $y = \begin{bmatrix} -9 \\ 2 \\ 3 \end{bmatrix}$ as linear combination of vector S. **04**

b. Given basis $S = \left\{ \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} \right\}$ for R^3.Use Gram–Schmidt process to transform S to an orthogonal basis for R^3. **05**

c. Find the least square solutions to the system $Ax = b$, where $A = \begin{bmatrix} 1 & -6 \\ 1 & -2 \\ 1 & 1 \\ 1 & 7 \end{bmatrix}$, $b = \begin{bmatrix} -1 \\ 2 \\ 1 \\ 6 \end{bmatrix}$. **05**